Digitalizing Midwifery Care: AR and VR Solutions for Maternal Education

Project Proposal Report

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BSc (Hons) in Information Technology – Interactive Media

Department of Information Technology

Sri Lankan Institute of Information Technology

June 2025

Automated Fetal Health Risk Assessment System for Rural Midwifery Care using Lightweight Machine Learning

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Declaration, copyright statement and the statement of the supervisor

We declare that this is our own work and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Mr. Didula Thanaweera Arachchi

Signature of the supervisor:

Date: 8/29/2025

Abstract

This research proposes a mobile application designed to enhance midwifery practice during the antenatal (before birth) and intrapartum (during birth) phases. The application integrates maternal health monitoring, decision-support tools, localized educational resources, and data-driven dashboards for both students and practicing midwives.

Current maternal care apps (such as Pregnancy and BabyCenter) are primarily designed for mothers and do not address midwives' competency development. Similarly, while training apps such as the Safe Delivery App provide clinical guidelines, they lack cultural adaptation, real-time monitoring, and data analysis features relevant to Sri Lanka.

The proposed system will include tools to record and track vital maternal indicators such as blood pressure, blood sugar, and fetal heart rate, and will provide decision-support modules for delivery scenarios such as breech birth, postpartum hemorrhage, and neonatal resuscitation. A data analysis dashboard will support midwives and students by visualizing trends, providing alerts, and offering feedback on performance.

The novelty of this project is its Sri Lankan cultural adaptation with Sinhala, Tamil, and English support, ensuring accessibility and adoption across rural and urban communities. By combining mobile app development with data analytics, this project aims to strengthen maternal safety and midwifery competency training in Sri Lanka.

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List of Abbreviations

Abbreviation	Description
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CTG	Cardiotocography (The technical term for the fetal heart rate monitoring trace).
FHR	Fetal Heart Rate (The main parameter analyzed by the AI).
ML	Machine Learning (The core technology used for prediction).
AI	Artificial Intelligence.
XAI	Explainable Artificial Intelligence (The feature that gives reasons for the risk).
VOG	Visiting Obstetrician and Gynaecologist (The specialist doctor).
SMOTE	Synthetic Minority Over-sampling Technique (Used to balance the dataset).
UCI	University of California, Irvine (The source of the dataset).
ASTV	Abnormal Short Term Variability (A key feature used by the model to detect risk).
API	Application Programming Interface (Used to connect the Python model with the Mobile App).
RF	Random Forest (The specific algorithm used).
MOH	Ministry of Health (Sri Lanka).
FHB	Family Health Bureau (Sri Lanka).

Introduction

Background & Literature survey

This research proposes an automated Machine Learning (ML) based component designed to assist midwives in the early detection of fetal health risks during the antenatal period. In Sri Lanka's rural healthcare sector, while Cardiotocography (CTG) machines are available in peripheral units, the expert knowledge required to interpret complex fetal heart rate patterns is often limited to Obstetricians (VOGs), who are not always available. This delay in expert interpretation can lead to preventable adverse outcomes such as fetal hypoxia, cerebral palsy, or stillbirths.

Current automated analysis systems, such as "SisPorto," are primarily desktop-based and expensive, making them unsuitable for mobile midwifery units. Similarly, existing Deep Learning models often require high computational power and active internet connectivity, restricting their use in low-resource rural settings. Furthermore, most AI models function as "black boxes," providing predictions without clinical reasoning, which limits their adoption by medical staff.

The proposed system addresses these gaps by utilizing a lightweight **Random Forest Classifier** to analyze CTG parameters (e.g., baseline heart rate, accelerations, decelerations) and instantly classify fetal health into **Normal, Suspect, or Pathological** categories. A key novelty of this project is the integration of **Explainable AI (XAI)**, which provides diagnostic reasons for high-risk predictions (e.g., "Prolonged Decelerations Detected"), thereby building trust and aiding midwives in decision-making. Designed to function **offline on standard mobile devices**, this system aims to bridge the specialist gap in rural healthcare, ensuring timely interventions and strengthening maternal and fetal safety across Sri Lanka.

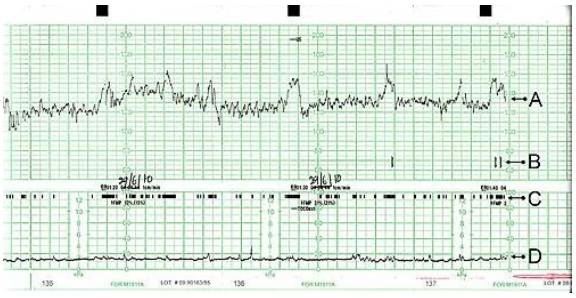


Figure 1 Cardiotocography Report

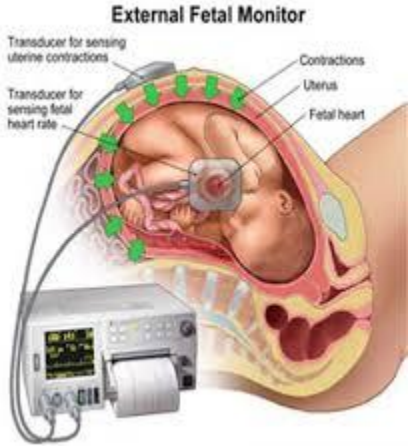


Figure 2 Cardiotocography

Research Gap

Several research studies on automated fetal health monitoring and Machine Learning (ML) applications have been conducted, but very few are directly applicable to **rural midwifery practice** in low-resource settings. Most existing automated systems are either designed for high-tech hospital environments (requiring expensive hardware) or rely on complex Deep Learning models that demand high computational power. While these systems achieve high accuracy, they lack the **portability, offline capability, and explainability** required for a mobile-based solution used by field midwives.

Research A – SisPorto 2.0 (Ayres-de-Campos et al., 2000) is a renowned computer-based system developed to analyze Cardiotocography (CTG) signals. It mimics expert interpretation with high clinical validity. However, it is a **desktop-based software** designed specifically for well-equipped hospitals with specialist doctors. It requires dedicated hardware and is not portable, making it unsuitable for rural midwives who conduct mobile clinics or home visits.

Research B – Deep Learning Approaches (e.g., Zhao et al., 2019) utilize Complex Neural Networks (CNNs) to classify fetal states with high precision. While scientifically advanced, these models suffer from high **computational load**, requiring powerful GPUs or cloud servers to function. Furthermore, they often operate as **"Black Boxes,"** providing a risk prediction without offering clinical reasons (explainability),

which reduces trust among medical staff who need to understand *why* a fetus is at risk.

The noticeable research gap is that while automated diagnostic tools exist, there are no comprehensive **mobile-based applications in Sri Lanka** that combine:

- **Lightweight Machine Learning:** Optimized algorithms (like Random Forest) capable of running **offline** on standard mobile smartphones without internet dependency.
- **Explainable AI (XAI):** Features that not only predict the risk but also provide **diagnostic reasoning** (e.g., identifying "Prolonged Decelerations") to support the midwife's decision-making.
- **Low-Cost Accessibility:** A solution designed specifically to bridge the specialist gap in **rural peripheral units** where VOGs are not immediately available.

Research Problem

Midwives in rural Sri Lankan hospitals face a critical **"Interpretation Gap"** when monitoring fetal wellbeing. While Cardiotocography (CTG) machines are available in many peripheral units, the advanced clinical expertise required to interpret complex fetal heart rate patterns (such as subtle decelerations or variability changes) is typically limited to Obstetricians (VOGs). In rural settings, a VOG is not always physically present, especially during night shifts or in emergencies.

Current practices rely heavily on the midwife's manual observation, which can be subjective and prone to human error. Existing automated analysis systems are restricted to desktop computers in major hospitals and are not portable. Consequently, there is a lack of digital tools that provide **instant, expert-level diagnostic support** on mobile devices. This delay in accurate diagnosis often leads to preventable adverse outcomes, such as fetal hypoxia, cerebral palsy, or stillbirths, due to the inability to detect distress signals in real-time.

Problem Statement

How can a lightweight, offline-capable Machine Learning model be engineered to automate the risk assessment of fetal health using CTG parameters, providing real-time, explainable diagnostic support for rural midwives in Sri Lanka?

Author / Existing System	**Methodology / Focus**	**Research Gap (Limitations)**
Ayres-de-Campos et al. (2000) *(The SisPorto 2.0 System)*	Developed a computer-based system to analyze CTG signals in hospitals, achieving high clinical accuracy acceptable to expert obstetricians.	**Desktop-based & Expensive:** Designed for high-tech hospitals with dedicated hardware. It is not portable and requires a specialist doctor to operate, making it unsuitable for rural mobile midwifery units.
Zhao et al. (2019) *(Deep*	Utilized Complex Neural Networks (CNN) and Deep Learning algorithms to	**High Computational Load:** These models require powerful GPUs or cloud servers. They are too "heavy" to run

Author / Existing System	Methodology / Focus	Research Gap (Limitations)
Learning on CTG)	classify fetal states with high precision.	offline on standard mobile phones used by field midwives and depend on active internet connectivity.
Ogasawara et al. (2021) *(Machine Learning Approaches)*	Applied standard ML models like SVM for fetal heart rate classification to automate diagnosis.	**"Black-Box" Issue:** These models provide a prediction (e.g., "Pathological") but fail to provide **Explainability** (clinical reasons). This lack of transparency reduces trust among medical staff who need to know *why* a fetus is at risk.
Maternity Foundation (2021) *(Safe Delivery App)*	A mobile app providing clinical guidelines and instructional videos for midwives to handle birth complications.	**Static Content Only:** While excellent for education, it lacks **Real-time Diagnostic Capabilities**. It cannot analyze patient data (CTG) or provide instant risk predictions for a specific mother.
Proposed System *(Automated Risk Assessment)*	**Lightweight Random Forest Classifier + Explainable AI (XAI)** running on a mobile architecture.	**Fills the Gap:** Provides a comprehensive solution that is **Mobile-First, Offline-Capable**, and **Explainable**. It brings expert-level diagnostics to low-resource rural settings without needing heavy hardware.

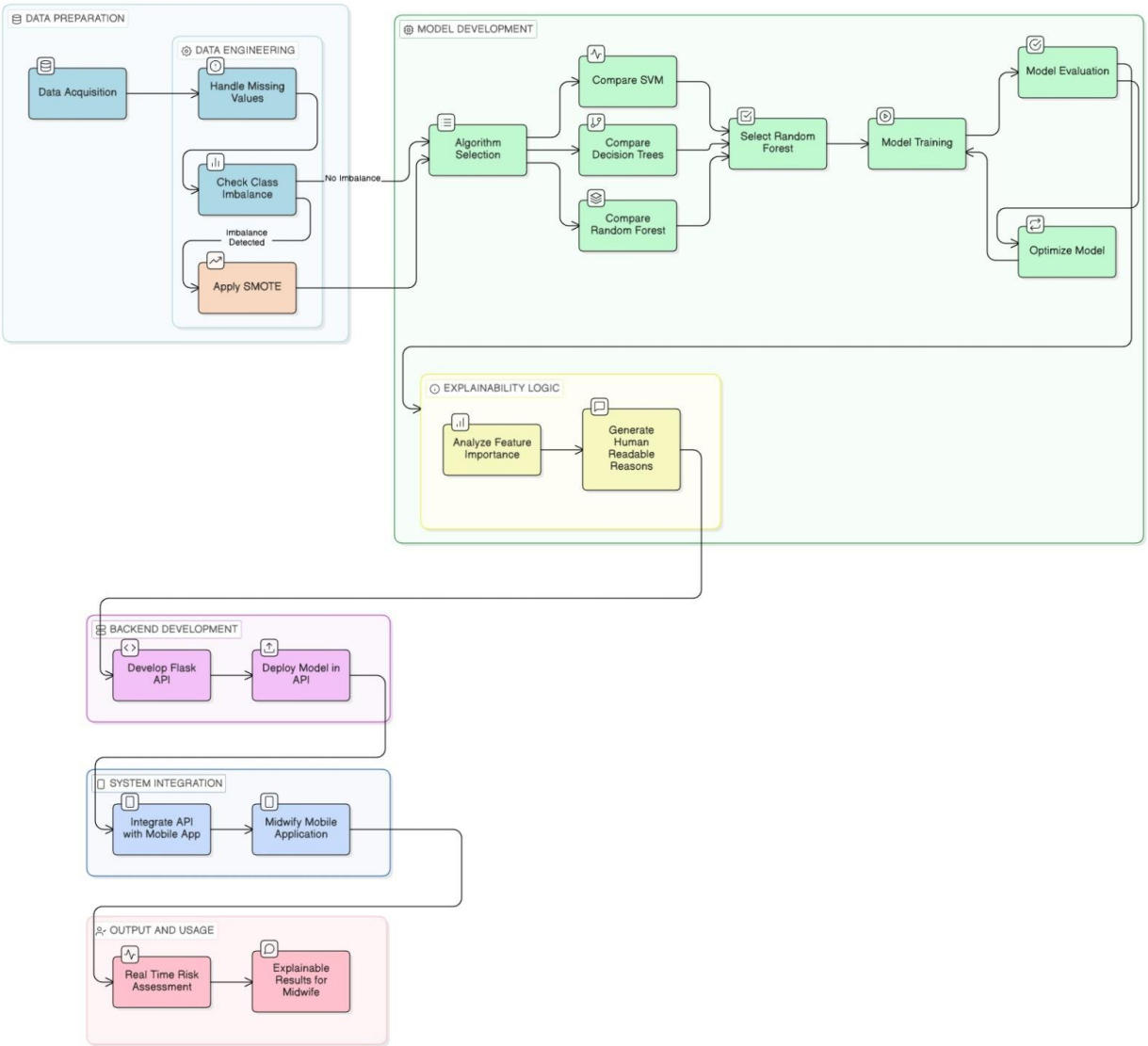
Objectives

Main Objectives

To engineer an automated, offline-capable Machine Learning component that classifies fetal health risks based on CTG parameters, providing real-time, explainable diagnostic support for rural midwives.

Specific Objectives

Figure 1 System Diagram



1. To analyze and preprocess clinical CTG data using **SMOTE** (Synthetic Minority Over-sampling Technique) to handle class imbalance and ensure high sensitivity in detecting pathological cases.
2. To train and optimize a lightweight **Random Forest Classifier** model capable of performing accurate risk assessment on mobile devices with limited computational resources.
3. To implement **Explainable AI (XAI)** logic to extract and display diagnostic reasons (e.g., "Prolonged Decelerations") for risk predictions, enhancing clinical trust.
4. To develop a secure **RESTful API** using Python Flask to integrate the predictive model with the main mobile application for real-time inference.
5. To validate the model's performance using standard evaluation metrics (Confusion Matrix, Recall, Precision) against the verified UCI clinical dataset.

Machine Learning Pipeline Development

- Data Engineering: Preprocessing using Python Pandas and Scikit-learn.
- Model Optimization: Hyperparameter tuning for lightweight performance.
- Explainability: Feature importance extraction for clinical reasoning.

Backend API Integration

- API Architecture: Python Flask microservice for model hosting.
- Communication: JSON-based data exchange between Mobile App and ML Model.
- Deployment: Offline-ready architecture for low-resource settings.

Methodology

The system development involves:

The development of the AI component follows the standard Data Science Lifecycle (DSLC):

1. **Data Acquisition & Analysis** – Retrieving the Fetal Health Dataset from the UCI Machine Learning Repository and analyzing clinical features.
2. **Data Preprocessing** – Cleaning data, handling missing values, and applying **SMOTE (Synthetic Minority Over-sampling Technique)** to balance the dataset for accurate risk detection.
3. **Model Engineering** – Training and optimizing the **Random Forest Classifier** to ensure high accuracy while remaining lightweight for mobile use.
4. **Backend API Development** – Creating a Python Flask API to host the trained model and enabling communication with the mobile app.
5. **Testing & Validation** – Validating the model using Confusion Matrices and conducting user acceptance testing (UAT) with potential users.

Table 2 Technologies Used

Technology	***Description***
Python (Pandas, Scikit-learn)	*Core programming language and libraries used for data preprocessing, machine learning model training, and evaluation.*
Google Colab	*Cloud-based development environment used for high-performance model training and experimentation.*
Random Forest Classifier	*The specific Machine Learning algorithm selected for fetal risk classification due to its high accuracy and low computational cost.*
Flask (Python Framework)	*Lightweight web framework used to create the RESTful API that connects the AI model with the mobile application.*
Pickle (.pkl)	*Serialization format used to save the trained model for efficient loading and offline inference.*
VS Code	*Integrated Development Environment (IDE) used for backend coding and API debugging.*
Postman	*Tool used for testing API endpoints and verifying JSON request/response structures.*

Functional Requirements

1. **Data Input:** The system must accept 21 CTG parameters (including Baseline Heart Rate, Accelerations, Uterine Contractions) as input.
2. **Risk Prediction:** The system must accurately classify the fetal state into three categories: **Normal, Suspect, or Pathological**.
3. **Explainability:** The system must identify and display the specific reasons for a high-risk prediction (e.g., "Prolonged Decelerations Detected").
4. **Confidence Scoring:** The system should provide a probability score (percentage) indicating the confidence level of the prediction.
5. **Offline Capability:** The prediction engine must function locally on the device or via a local server without requiring active internet connectivity.

Non-Functional Requirements

1. **Accuracy:** The AI model must achieve a testing accuracy of **90% or higher** on the validation dataset.
2. **Performance:** The risk prediction result must be generated within **2 seconds** after data submission.
3. **Reliability:** The system must provide consistent results for the same input values (Deterministic behavior).

4. **Resource Efficiency:** The trained model file size should be optimized (lightweight) to ensure it runs smoothly on mobile devices with limited RAM.
5. **Scalability:** The backend API should be able to handle multiple concurrent requests without crashing.

System Requirements

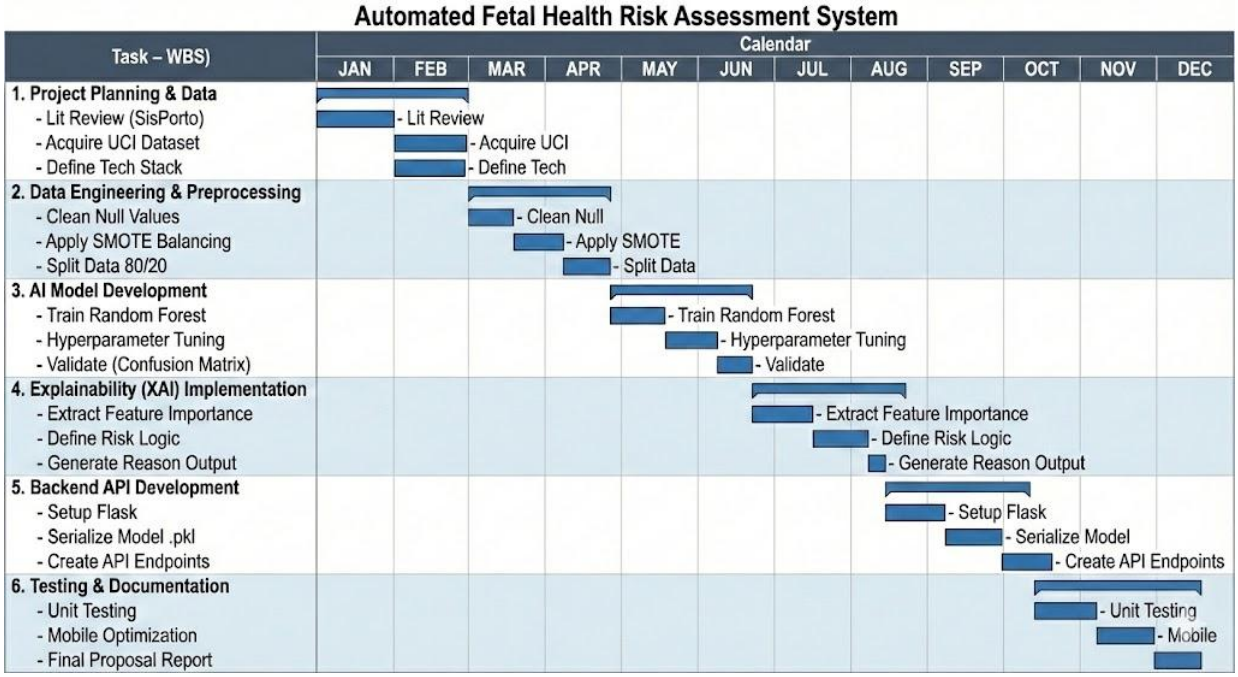
Hardware Requirement

1. Development: Laptop with at least 8GB RAM (for running ML training locally if needed) or access to Cloud GPUs (Google Colab).
2. Testing: Android Smartphone (Mid-range) to test the mobile integration and response speed.

Software Requirements

1. **Language:** Python 3.8 or higher.
2. **Libraries:** Scikit-learn, Pandas, NumPy, Flask.
3. **Tools:** Visual Studio Code, Postman, Git.

Figure 4 Gantt Chart



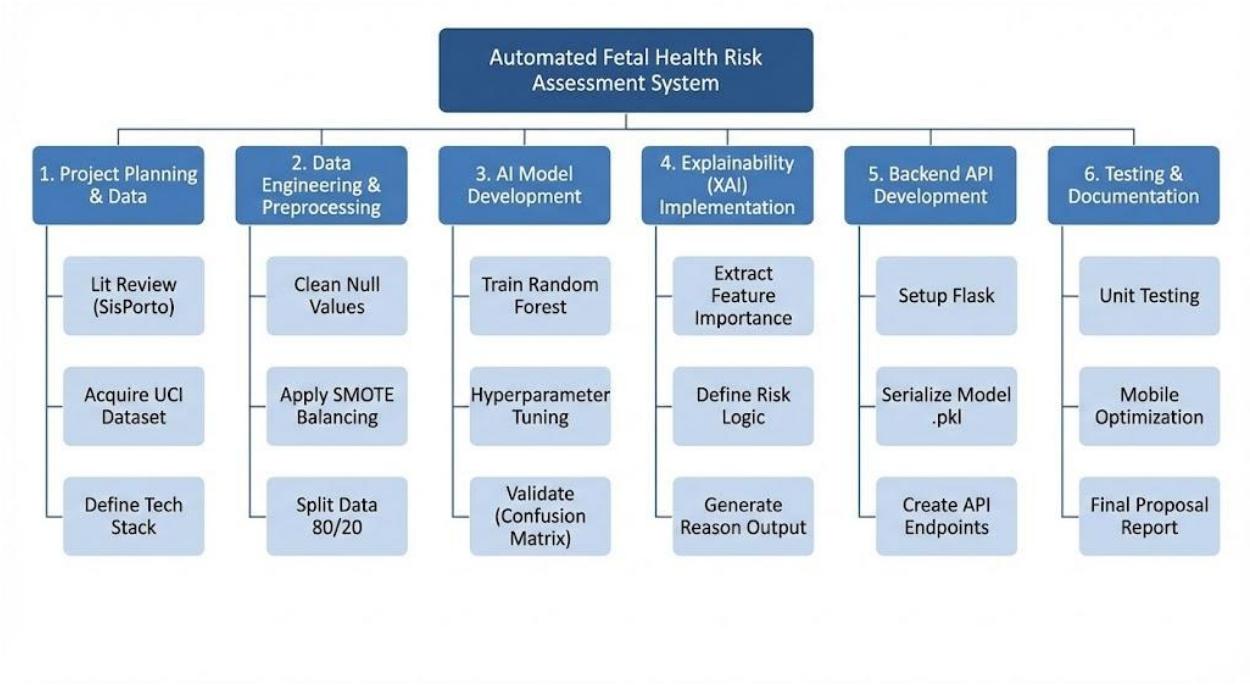


Figure 5 Work Breakdown Structure (WBS)

Description of Personal and Facilities

Supervisors

Our supervisor Mr. Didula Thanaweera Araghchi and Co-Supervisor Mrs. Narmada Gamage, who are experts in the technical field, will provide guidance in software development and technology integration.

External Supervisor

Our external supervisor is a midwife who has been in the field for over 26 years. She shared real experiences, how she manages mothers before, during and after birth. Her knowledge helped us understand the real problems midwives face daily. With her guidance, we were able to plan the app feature more practically and usefully.

Facilities

The project is carried out with support from the Humanities and Nursing Departments of SLIIT. Access is provided to medical books, maternal health guidelines, and computer labs for app development and testing. This setup ensures that the mobile app is developed in line with real-world midwifery practices in Sri Lanka.

Sample of Midwifery Students

A group of 15–20 midwifery and nursing students will take part in the testing phase. They will use the app, give feedback on its features, and help improve its design and content. This hands-on testing also gives students practical learning opportunities.

Device Access

The app will be tested on commonly used Android smartphones, including low and mid-range devices. This ensures it works well for public health midwives in both urban and rural areas. The testing will also cover local language support, offline access, and data entry functions.

Budget and Budget Justification

Table 3 Budget Table

Expenses	
Requirement	Cost (Rs.)
Mobile Devices for Testing (Android + iOS)	180,000
Backend Server Hosting (1 year)	60,000
Firebase / Cloud Database (annual)	45,000
Software Licenses (Figma, IDEs, API tools)	30,000
Data Analysis Tools (Python Libraries, Power BI Pro)	50,000

Reference List

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Digitizing Midwifery Care: AR and VR Solutions for Maternal Education

25-26J-426

Project Proposal Report

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BSc (Hons) in Information Technology – Interactive Media

Department of Information Technology

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August 2025

Digitizing Midwifery Care: AR and VR Solutions for Maternal Education

25-26J-426

Project Proposal Report

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Co-Supervisor - Mrs. Narmada Gamage

BSc (Hons) in Information Technology – Interactive Media

Department of Information Technology

Sri Lankan Institute of Information Technology

August 2025

Declaration page of the candidates & supervisor

We declare that this is our own work and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning, and to the best of our knowledge and belief, it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for their undergraduate dissertations under my supervision.

Mr. Didula Thanaweera Arachchi

Signature of the supervisor: Date: 07/09/2024

Abstract

Midwives play a pivotal role in early childhood health across Sri Lanka, especially in rural and underserved communities. However, they often lack access to digital tools that help identify infant cranial and postural abnormalities early. This research proposes the development of a bilingual (Sinhala/English) mobile application that integrates an AI-assisted infant cranial and posture monitoring module. Using computer vision techniques, the app will analyze infant images or videos to detect cranial deformities and postural irregularities, providing actionable recommendations for early intervention. A bilingual interface will ensure accessibility for Midwives across diverse literacy and language backgrounds. The project aims to evaluate both the technical accuracy of the AI module and the usability of the bilingual interface with midwives in Sri Lanka.

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List of abbreviations

AI	Artificial Intelligence
CNN	Convolutional Neural Network
CVAI	Cranial Vault Asymmetry Index
CI	Cephalic Index
mHealth	Mobile Health
UI	User Interface
UX	User Experience
SUS	System Usability Scale
IDE	Integrated Development Environment
API	Application Programming Interface
SQLite	Structured Query Language Lite
SDK	Software Development Kit
BaaS	Backend as a Service
FOSS	Free and Open-Source Software
LKR	Sri Lankan Rupees
NLP	Natural Language Processing
RGB	Red, Green, Blue (color model used in image processing)

ROI	Region of Interest (used in image analysis)
FPS	Frames Per Second
UI/UX	User Interface / User Experience
ML	Machine Learning
API	Application Programming Interface

Introduction

This research project will create and test a mobile phone app that helps midwives in Sri Lanka find early signs of head shape problems and body posture issues in babies. The app will use computer tools called TensorFlow and OpenCV to look at pictures or videos of infants and give important health numbers, such as CVAI (which shows if the head is flat on one side), CI (which shows the head shape), and posture angles (which show how the baby's body is positioned). The app will be built using Flutter and will use Firebase to store and manage data safely. It will support both the Sinhala and English languages so that midwives from different areas can use it easily, even if they speak different languages or have different reading skills. The project will begin with reading many research papers and preparing a dataset of baby images. Then, the team will build the AI system, design the mobile app, and connect the AI to the app so it can work in real time or with saved images. Midwives will test the app and give feedback using a score system called the System Usability Scale (SUS), and they will also share their thoughts to help improve the design. The AI system will be tested to see how correct and reliable it is by comparing it with expert standards. The final goal is to make a tool that fits the culture and needs of Sri Lankan midwives, helps them give better care, and improves the health of babies.

Background & Literature survey

In Sri Lanka, midwives are the main caregivers for mothers and babies, especially in villages and places with fewer resources. They play a very important role in keeping mothers and infants healthy. But many midwives do not have digital tools to help them find early signs of health problems in babies. Some babies may have head shape problems like plagiocephaly (flat head on one side) or brachycephaly (short and wide head), or body posture issues. If these problems are not found early, they can cause long-term health issues. It is very important to find these problems early and give care

quickly. But right now, most ways to check babies are done by hand, depend on personal judgment, or require special machines that are not available in poor areas. This research project wants to solve that problem by making a mobile app that uses AI (artificial intelligence) to help midwives check babies' head shapes and body posture. The app will work in both Sinhala and English so that midwives with different reading and language skills can use it easily.

New mobile health technologies have shown that it is possible to use computer vision and machine learning to study baby pictures and give useful health information. For example, Nakahari and others tested an app called "Measurement of Baby's Head Shape™." This app uses smartphone pictures to calculate CVAI (Cranial Vault Asymmetry Index) and CI (Cephalic Index), which are numbers that show head shape. The app worked very well, with high accuracy and good results. But it was only tested on plastic head models, not on real babies. So, it still needs more testing in real-life situations, where babies move, lighting changes, and hair may cover parts of the head.

Another study by Neves and others created a tool called PosturAll. It looks at still pictures and gives scores to show the risk of body posture problems. It worked well and gave results **like** professional tools. But it was tested only on adult students, not on children. So, it may not work the same way for babies. Still, the tool works on mobile phones and uses smart ways to find body markers, which means it could be changed to help with child health.

Tabatabaei and others made an app called INCROS. It uses deep learning to find head shape problems from smartphone pictures. The app uses a smart model called U-Net with ResNet-18 and gives results in just 11 seconds. It worked very well, with 99.01% accuracy. The app is easy to use caregivers only need a tight cap and a coin for size reference. But the app was tested on a small number of pictures, and it still needs to be tested in real-world settings.

Some researchers have also used AI to study signals from the baby's body. León and others made a model that uses heart rate, breathing, and other signals to guess the baby's

developmental age. The model was very accurate and helped find delays in very small or early born babies. This shows that AI can help with baby health in ways that do not need touching or special machines.

Shamsir and others went even further. They made a system with many sensors and AI to find problems like breathing stops (apnea), seizures, and discomfort. It worked very well with fake data and gave alerts in real time using mobile and cloud systems. But because it used fake data and was not tested in hospitals, it cannot be used in clinics yet. Still, it shows how sensors and AI can work together to care for babies.

In the area of mothers' health, wearable tools and apps that use two languages have helped people make better health choices. Zhang and others made a wearable system that uses AI to study labor progress and decide when a mother should go to the hospital. It worked well and was tested against hospital machines. But it only worked in one language, so it was hard for people who speak other languages to use it.

AI has also been used in talking systems. Al-Mutairi and others made a bilingual chatbot that helps with school advice in English and Arabic. It was not made for health, but it's designed to separate AI systems for each language, and as a backup system when the answer is not clear, it can be used in health apps too. But the Arabic part had problems like no spell check and weak grammar tools, which show that language tools need to be better for health use.

All these studies show that AI-powered mobile apps can change how we care for mothers and babies. They show that it is important to make tools that are easy to use, accurate, and available to everyone, especially in places with fewer resources. Your research project builds on these ideas. You are making a bilingual mobile app for midwives in Sri Lanka. It will use computer vision to study baby head shapes and posture. The app will be designed for people with different reading levels and language skills. This mix of smart technology and kind design can help midwives give better care and improve the health of babies across the country.

Research Gap

One big problem in checking baby health is that there are not enough easy-to-use and trusted tools to find head shape problems in real-life places like homes or small clinics. Some apps, like Measurement of Baby's Head Shape™ (by Nakahari) and INCROS (by Tabatabaei), show good results in finding head shape issues like plagiocephaly (flat head on one side) and brachycephaly (short and wide head). But these apps were tested in special, controlled places. Nakahari's app was tested only on plastic head models, not real babies. INCROS needed a tight cap and a coin to help measure the head correctly. Even though INCROS gave very accurate results—more than 99%, it was only tested on 28 baby pictures. These limits make it hard to trust apps in real-life places like village clinics or homes. To solve this, my research will create a mobile app that works in both Sinhala and English. It will use AI (artificial intelligence) to find head shape problems using real baby pictures taken in everyday settings. The app will be tested with midwives in Sri Lanka to make sure it works well and is easy to use in places with fewer resources. Another big gap is that there are no good tools made just for checking baby posture. One tool called PosturAll (by Neves) uses pictures and smart computer methods to check body posture. It gave good results—over 92% accuracy. But it was only tested on adult students, not babies. Babies move and grow differently from adults, so the tool may not work well for them. My research will add posture checking to the mobile app using pictures and data made just for babies. The AI system will be trained using baby-specific images. Midwives will test the app to make sure it fits well into their daily work and helps them care for babies better.

Language is also a big barrier in health technology, especially in places like Sri Lanka, where people speak different languages. One example is a bilingual chatbot made by Al-Mutairi. It used English and Arabic and showed that it is possible to build systems that understand more than one language. But this chatbot was made for school advice, not health care. It only gave 75-80% correct answers, which is not good enough for health use. Also, the Arabic part had problems like no spell-check and weak grammar tools,

which can be risky in health settings. My research will create a Sinhala and English interface made specially for midwives. It will use clear health words and smart language tools to answer common questions and give guidance. If the app is not sure about a question, it will show helpful pictures or messages to keep things safe and easy to understand.

Lastly, there are very few tools that can watch both mothers and babies in real time and are also easy to use for different cultures and languages. Some systems, like the wearable labor monitor (by Zhang) and the smart baby monitor (by Shamsir), use sensors and AI to check body signals and send alerts to phones. These systems worked well—over 96% and 99.88% accurately—but they were tested only on small groups or fake data. They did not support more than one language, and their designs were not made for people from different backgrounds. They also did not have chat support to help users. My research will fill this gap by making a mobile app that checks a baby's head shape and posture using AI and includes a bilingual chat system. The app will protect privacy, give alerts in real time, and offer advice that fits the local culture. This will help midwives and families in Sri Lanka take better care of babies using one simple, inclusive digital tool.

Table 1 Comparative Table

Area	**Existing Studies / Tools**	**Strengths**	**Gaps**
Head Shape Analysis	*Measurement of Baby's Head Shape*™ (Nakahari) *INCROS* (Tabatabaei)	- Automated CVAI & CI calculation - High accuracy (>99%)	- Tested on models or small datasets - Needed caps/coins - Lab conditions only
Posture Analysis	*PosturAll* (Neves)	- Accurate posture risk scoring (>92%)	- Tested on adults, not babies

Real-World Testing	Multiple studies	- Showed proof of concept	- No field testing with health workers
Bilingual Support	*Al-Mutairi's Chatbot*	- Multi-language design	- Low accuracy (75–80%) - No health focus
Mother & Baby Monitoring	*Labor Monitor* (Zhang) *Smart Baby Monitor* (Shamsir)	- High accuracy, real-time alerts	- Small samples, simulated data - No language/cultural support

Research Problem

In Sri Lanka, midwives are very important for checking the health and growth of babies, especially in villages and places with fewer resources. But many midwives do not have digital tools to help them find early signs of problems in a baby's head shape or body posture. These problems, like flat head or poor posture, can cause serious health issues later if they are not found and treated early. Right now, most ways to check these problems are done by hand, depending on personal judgment, or require special machines that are not available in small clinics or homes. Also, many health apps do not support the Sinhala language, so Sinhala-speaking midwives cannot use them easily. This research aims to solve both problems: the lack of accurate tools and the lack of language support. It will create a mobile app that works in both Sinhala and English. The app will use AI (artificial intelligence) to study pictures and videos of babies and find head shape and posture problems. It will use computer vision to do this and show the results in a simple and culturally friendly way. By mixing smart technology with kind design, this project hopes to give midwives a tool that is both medically correct and easy to use, helping them take better care of babies all across Sri Lanka.

Objectives

This research project has clear goals. It wants to help midwives in Sri Lanka by giving them a smart and easy-to-use tool to find early signs of problems in a baby's head shape and body posture. In many places with fewer resources, old methods of checking babies are not available or do not work well. Also, many health apps do not support the Sinhala language and are not accurate enough for medical use. To fix these problems, this project will create a mobile app that works in both Sinhala and English. The app will use artificial intelligence (AI) to study pictures or videos of babies and give simple, useful results. These goals will guide how the app is built, designed, and tested. The final app will be strong in science, kind in design, and easy for midwives to use in different places and languages.

Main Objectives

The main goal of this research is to create and test a mobile app that works in both Sinhala and English. This app will help midwives in Sri Lanka check babies for early signs of head shape problems and posture issues. Many midwives, especially in villages and low-resource areas, do not have easy tools to do these checks. This app will use smart computer methods called AI and computer vision to study pictures or videos of babies. It will give important health figures like CVAI (which shows if the head is flat on one side), CI (which shows the head shape), and posture angles (which show how the baby's body is positioned). The app will be built using Flutter and Firebase and will work in both Sinhala and English so that midwives with different reading levels and language backgrounds can use it easily. To make the app more **interactive and supportive**, the **output will give audio and video guidance** based on the baby's assessment. The app will be tested to make sure it is accurate, easy to use, and fits well with the way midwives work. The final goal is to give midwives a kind and smart tool that helps them care for babies better and improves early health outcomes across Sri Lanka.

Specific Objectives

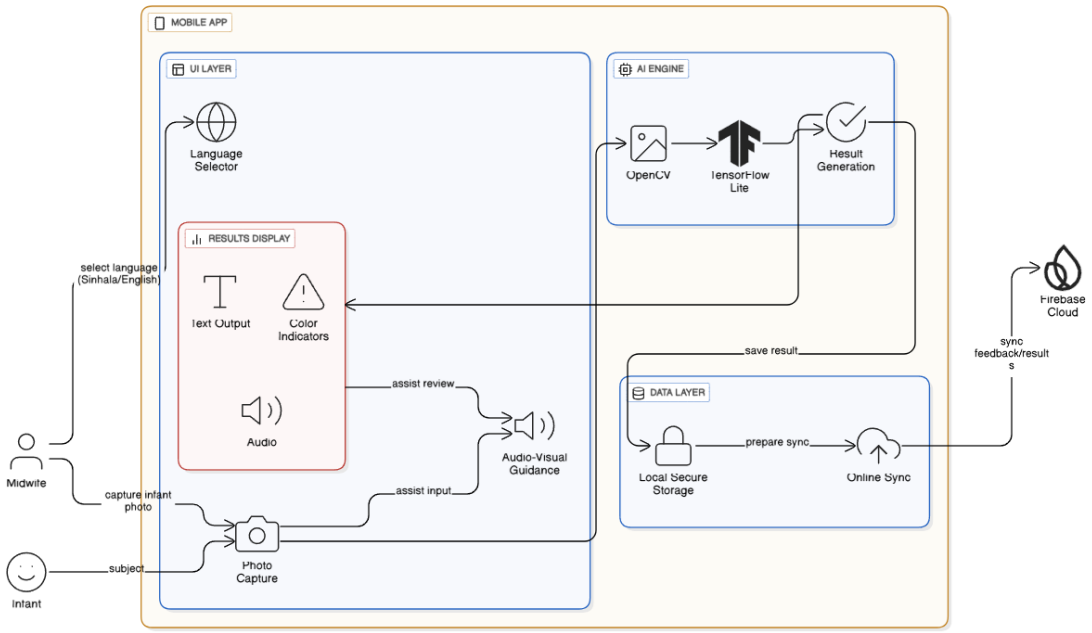
Specific objectives are small, clear steps that help guide a research project. While the main goal gives the big picture, specific objectives break it down into smaller tasks that are easier to plan and measure. In this project, the specific objectives focus on both the technical side and the user experience. The aim is to build a mobile app that uses AI to help midwives in Sri Lanka check babies for head shape and posture problems. Each objective helps make sure the app is not only medically correct but also easy to use, kind in design, and helpful for midwives in different places and languages.

1. Design and build an AI system to find head shape and posture problems from baby pictures or videos. - This step will create a smart computer model using tools called TensorFlow and OpenCV. The model will study images or videos of babies and give important health numbers like CVAI (shows if the head is flat), CI (shows head shape), and posture angles (shows how the baby's body is positioned). This will help find problems early without needing special machines.
2. Make a mobile app that works in Sinhala and English. - The app will be built using Flutter, which works on a variety of phones, and Firebase, which helps store and manage data. It will have a bilingual interface so midwives who speak Sinhala or English can use it easily.
3. Integrate audio-visual storytelling to improve understanding and interactivity. - The app will play short, pre-recorded Sinhala/English audio narrations and display simple looping motion graphics that explain the detected result and recommended action. This helps midwives, including those with lower literacy, interpret results quickly and confidently.
4. Connect the AI system to the app and only show results. - This step will put the trained AI system inside the app. It will let midwives check baby pictures or videos in real time or later. The results will be shown in a simple, color-coded

format with easy language so midwives can understand and act quickly, even without medical training.

5. Test the app with midwives to see if it is easy and friendly to use. - Midwives will try the app and give feedback. A score system called SUS (System Usability Scale) and interviews will be used to see how easy and helpful the app is. This will make sure the app feels supportive and fits well with how midwives work in both cities and villages.
6. Check how accurate and reliable the AI system is. - The AI system will be tested by comparing its results with expert opinions or trusted medical tools. Numbers like accuracy, precision, recall, and F1-score will be used to see how well the system works and if it is ready to be used in real life.

Figure 1 System Diagram



Methodology

The project will begin with a detailed literature review to study past research on mobile health applications, tools for cranial deformity and posture assessment, and bilingual health technologies. Special focus will be given to solutions that use AI for infant health screening and user interface designs that work well in low-resource settings. This review will ensure the proposed app builds on proven ideas, addresses existing gaps, and is appropriate for midwives working in rural and urban Sri Lanka.

Next, the project will focus on data collection. Publicly available cranial and posture datasets will be used first. If ethically approved, a small local dataset will be created using images or videos of Sri Lankan infants collected in homes or clinics. Parents or guardians will provide informed consent, and all images will be anonymized and securely stored. This approach ensures that the AI system learns from real-world data that reflects the conditions midwives face every day while respecting privacy and cultural norms.

After data collection, AI model development will begin. Convolutional Neural Networks (CNNs) will be built using TensorFlow and OpenCV to detect cranial abnormalities such as CVAI and CI, as well as posture issues by analyzing angles and positions. Data augmentation techniques will be applied to make the model robust under different lighting and infant positions. The trained model will then be optimized using TensorFlow Lite to run efficiently on entry-level Android devices, ensuring the solution is accessible even in low-resource settings.

Once the AI model is ready, a bilingual mobile application will be developed using Flutter so that it works on multiple platforms. Firebase will be used for cloud data storage and authentication. The user interface will be simple and intuitive, featuring culturally relevant icons and bilingual (Sinhala/English) support to make the system accessible for midwives with varying language skills and technical familiarity.

The AI module will be integrated into the mobile app so that midwives can upload images or videos and receive results in real time or after processing. Results will be displayed in a clear, color-coded format (green for normal, yellow for monitor, red for refer) along with short, easy-to-follow recommendations for action. This integration will be tested on different devices and in different environments to ensure smooth performance in rural and urban settings.

To make the system more engaging and easier to understand, audio-visual storytelling will be integrated into the app. Short pre-recorded voice narrations in Sinhala and English will explain the results, and simple looping motion graphics will visually demonstrate the detected condition and recommended next steps. This feature will help midwives, including those with limited literacy, to understand the outcomes clearly and take appropriate action confidently.

Finally, evaluation and usability testing will be conducted. The AI model's technical performance will be measured using accuracy, precision, recall, and F1 score, comparing its predictions against expert medical evaluations. Usability testing will be carried out with midwives in real-life scenarios, using the System Usability Scale (SUS) and interviews to gather feedback on ease of use, language clarity, and cultural relevance. The feedback collected will guide refinements to improve the final prototype and provide recommendations for scaling the system across Sri Lanka.

Functional Requirements

Functional requirements describe what your system must do—the core features that enable midwives to use the app effectively in real-world conditions.

1. User Authentication and Access - This ensures that only authorized users (e.g., midwives) can access the app. It protects sensitive health data and allows for future role-based access, such as adding supervisors or researchers.

2. Image and Video Capture - Midwives must be able to capture or upload images/videos of infants directly through the app. This feature is central to enabling AI analysis and must include visual guidance (e.g., framing grids) to ensure diagnostic accuracy.
3. AI-Powered Analysis - The app must process visual input using the embedded AI model to detect cranial deformities and postural irregularities. It should output clinical metrics like CVAI, CI, and posture angles, along with easy-to-understand visual indicators (e.g., color-coded risk levels).
4. Bilingual Interface - All content navigation, instructions, and results must be available in Sinhala and English. This ensures inclusiveness for midwives across linguistic backgrounds and supports training in both urban and rural contexts.
5. Result Interpretation and Guidance - AI-generated results must be presented in a simplified format, using culturally adapted language and visual cues. The app should offer actionable recommendations to support decision-making.
6. Audio-Visual Storytelling Integration – To improve understanding and interactivity, the app must play short, pre-recorded Sinhala/English audio narrations and display simple looping motion graphics that explain the detected result and recommended action. This helps midwives, including those with lower literacy, interpret results quickly and confidently.
7. Usability Feedback Collection - The app should include a built-in feedback module, allowing midwives to rate usability (e.g., SUS scores) and provide comments. This feedback loop supports iterative design and ensures the app evolves with user needs.
8. Data Privacy and Security - All personal and health data must be encrypted during storage and transmission. Users should have control over what data is stored, shared, or deleted, in compliance with Sri Lankan data protection laws.

Non-Functional Requirements

Non-functional requirements define how your system should behave - its performance, reliability, and ethical standards.

1. Performance - The app must be fast and responsive, even on entry-level Android devices. AI analysis should be completed within 10 seconds, and the interface should remain smooth during use.
2. Scalability - The system must be designed to grow, supporting more users, additional languages, and new health modules (e.g., nutrition tracking). Firebase should handle increased data volume without slowing down.
3. Usability and Accessibility - The interface must be intuitive for users with low digital literacy. This includes large buttons, simple language, culturally familiar icons, and minimal text input. Accessibility features like voice prompts or visual cues should be considered for future versions.
4. Maintainability - The codebase should follow clean architecture principles, making it easy to update, debug, and expand. Documentation should be provided for developers and health administrators.
5. Localization and Internationalization - The app must support Sinhala and English, with potential for future language additions (e.g., Tamil). Text layout, font rendering, and UI spacing should adapt to language-specific formatting and reading direction.
6. Security and Compliance - The system must comply with ethical standards and local regulations for handling health data. This includes secure authentication, encrypted storage, and audit trails for data access. Consent mechanisms must be built into any data collection process.
7. Ethical Design - AI outputs must be presented with disclaimers to avoid overdiagnosis or misinterpretation. The app should reinforce that it supports—not

replaces—clinical judgment. Emotional design principles should be applied to reduce anxiety and build trust among users.

System Requirements

System requirements define the technical specifications needed to develop, deploy, and operate a software solution effectively. They ensure that the application runs smoothly, remains secure, and performs reliably across different environments. These requirements are typically divided into two categories:

1. Hardware Requirements: The physical devices and infrastructure needed (e.g., smartphones, computers).
2. Software Requirements: The platforms, frameworks, and tools used to build and run the application.

For your project, system requirements ensure that midwives can use the app in low-resource settings and that developers have the right tools to build and maintain it.

Hardware Requirements

1. Mobile Devices - Android 8.0 or higher, minimum 2GB RAM, 16GB storage → For midwives to run the app in field conditions.
2. Development Machine - Windows/macOS/Linux, 8GB RAM, 256GB SSD, optional GPU → For AI model training, app development, and testing.
3. Test Devices - Multiple entry-level Android smartphones → To validate usability, performance, and compatibility across real-world devices.
4. Internet Access - Wi-Fi or mobile data (3G/4G) → Required for Firebase sync, updates, and remote monitoring.

Software Requirements

1. React Native → For cross-platform mobile app development (Android/iOS) and building the bilingual interface.

2. Node.js → For backend services, handling API requests, and managing data securely.
3. SQLite → For local data storage, storing app settings, results, and temporary data on the device.
4. TensorFlow Lite → For running the trained AI model on mobile to detect cranial deformities and posture issues.
5. Firebase → For authentication, cloud storage of images/videos, database management, and syncing data.
6. Android Studio → For building, testing, debugging, and deploying the mobile application on Android devices.
7. Audacity → For recording and editing audio guidance to support interactive output.
8. Adobe After Effects → For creating looping motion graphics that visually explain results and recommended actions.
9. Figma → For designing and prototyping the bilingual user interface.

Table 2 Technologies Used

Technology	**Purpose**	**Platform/Type**
React Native	Build cross-platform mobile app and bilingual interface	Mobile (Android/iOS)
Node.js	Backend services, API handling, data management	Server-side / Web
SQLite	Local storage for app settings, results, and temporary data	Mobile (Embedded Database)
TensorFlow Lite	Run AI model on device for cranial and posture detection	Mobile (On-device AI)

Firebase	Authentication, cloud storage, database management, data syncing	Cloud / Backend-as-a-Service
Android Studio	Build, test, debug, and deploy the app	Development Environment (IDE)
Audacity	Record and edit audio guidance to support interactive output	Audio Editing / Desktop Tool
Adobe After Effects	Create looping motion graphics that visually explain results and actions	Animation / Desktop Tool
Figma	Design and prototype the bilingual user interface	Design / Prototyping Tool

Figure 2 Gantt Chart

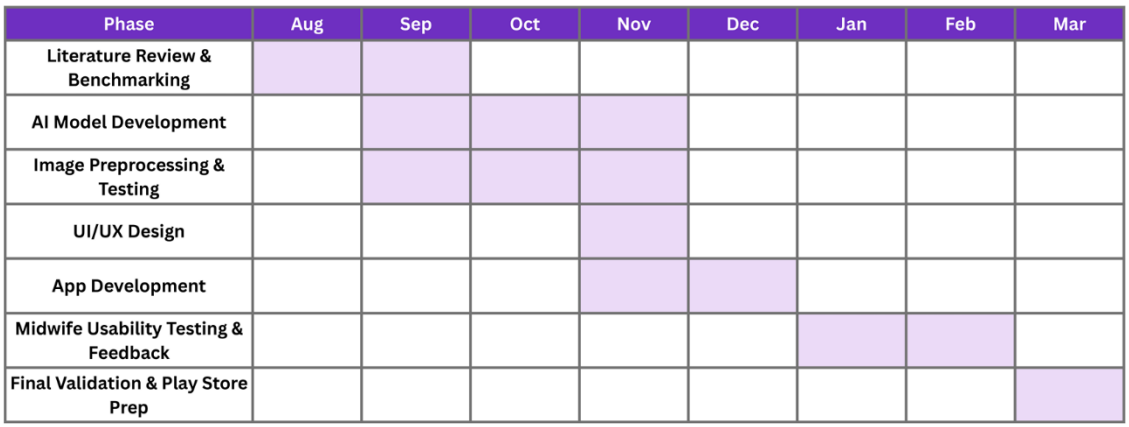
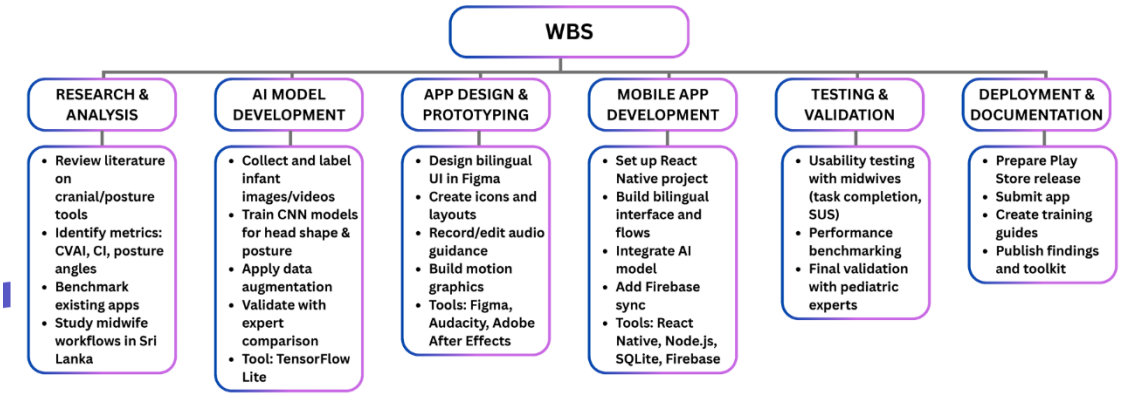


Figure 2 Work Breakdown Structure (WBS)



Description of Personal and Facilities

Supervisor

Our supervisor, Mr. Didula Thanaweera Araghchi, and Co-Supervisor Mrs. Narmada Gamage, who are experts in the technical field, will guide software development and technology integration.

External Supervisor

Our external supervisor is a midwife who has been in the field for over 26 years. She shared real experiences, how she managed mothers before, during, and after birth. Her knowledge helped us understand the real problems midwives face daily. With her guidance, we were able to plan the app feature more practically and usefully.

Facilities

The project is carried out with support from the Humanities and Nursing Departments of SLIIT. Access is provided to medical books, maternal health guidelines, and computer labs for app development and testing. This setup ensures that the mobile app is developed in line with real-world midwifery practices in Sri Lanka.

Device Access

The application will be tested on widely available Android smartphones, focusing on both entry-level and mid-range devices to ensure accessibility across different user groups. Testing will cover mothers, caregivers, and healthcare workers, ensuring smooth performance regardless of device specifications. In addition, the app will support local language interfaces, offline access for areas with poor connectivity, and secure data storage to allow continued use in diverse environments.

Budget and Budget Justification

This project requires a modest but strategically allocated budget to support the development, testing, and evaluation of a bilingual AI-assisted mobile application for infant cranial and posture monitoring. The budget prioritizes essential tools, infrastructure, and user engagement to ensure both technical feasibility and real-world usability.

Table 3 Budget Table

Item	**Estimated Cost (LKR)**	**Purpose**
Software Tools (TensorFlow, React Native, Figma, Adobe After Effects, Audacity)	15,000	For AI development, image processing, and UI/UX prototyping
Cloud Services (Firebase)	20,000	Backend infrastructure for secure data storage, authentication, and sync
Test Devices (Entry-Level Android Phones)	40,000	For performance testing and usability validation across real-world devices
Usability Testing (Compensation, Logistics)	15,000	Incentives and travel support for midwives participating in user studies
Miscellaneous / Contingency	15,000	Buffer for unexpected costs (e.g., translation, printing, accessories)
Total Estimated Budget	105,000 LKR	

Reference List

[1] Nakahari, T., Tabatabaei, S., & Hasegawa, Y. (2023). *Measurement of baby's head shape™: Evaluation of cranial deformity using plastic models. IEEE Xplore.* *Link*

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Digitalizing Midwifery Care: VR & Mobile Application Solutions for Maternal Education

Project Proposal Report

Kumarage M.B.S.I.E – IT22606556

BSc (Hons) in Information Technology – Interactive Media

Department of Information Technology

Sri Lankan Institute of Information Technology

June 2025

Machine Learning–Based Interactive Mobile App For Early Risk Detection and Midwifery Support

Project Proposal Report

Kumarage M.B.S.I.E – IT22606556

BSc (Hons) in Information Technology – Interactive Media

Department of Information Technology

Sri Lanka Institute of Information Technology

June 2025

Declaration, copyright statement and the statement of the supervisor

We declare that this is our own work and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
Kumarage M.B.S.I. E	IT22606556	

The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Mr. Didula Thanaweera Arachchi

Signature of the supervisor: Date: 8/22/2025

Abstract

This research proposes the design and development of an interactive mobile application that supports Sri Lankan midwives in antenatal and intrapartum care through real-time machine learning–based maternal risk prediction, interactive data visualization, and multimodal alerts. The system integrates maternal vital data such as blood pressure, blood sugar, fetal heart rate, and hemoglobin levels to predict early risk conditions including anemia and pregnancy-related complications.

Unlike existing maternal health applications that are either mother-focused or guideline-only tools, the proposed system is specifically designed for midwives and midwifery students. It provides real-time dashboards, explainable machine learning outputs, and voice and haptic feedback to improve decision-making speed, accuracy, and confidence. The application is culturally adapted with Sinhala, Tamil, and English language support and is optimized for low- and mid-range mobile devices commonly used in Sri Lanka. By combining mobile development, interactive media, and applied machine learning, this project aims to enhance maternal safety and strengthen midwifery competency.

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List of Abbreviations

Abbreviation	Description
mHealth	Mobile Health
BP	Blood Pressure
FHR	Fetal Heart Rate
ANC	Antenatal Care
PPH	Postpartum Hemorrhage
API	Application Programming Interface
UI	User Interface
UX	User Experience
SQL	Structured Query Language
DB	Database
ML	Machine Learning
WHO	World Health Organization
FHB	Family Health Bureau (Sri Lanka)
MOH	Ministry of Health (Sri Lanka)

Introduction

Midwives play a critical role in Sri Lanka's maternal healthcare system, particularly in antenatal monitoring and early detection of pregnancy-related risks. Although Sri Lanka has achieved significant reductions in maternal mortality, preventable complications such as anemia, hypertension, and delayed emergency response remain challenges, especially in rural settings.

With the increasing availability of digital health data, there is an opportunity to enhance midwives' decision-making using intelligent mobile tools. However, most existing research on maternal risk prediction using machine learning remains confined to academic models without real-time mobile deployment or integration into midwifery workflows. This project responds to that gap by proposing a machine learning–based interactive mobile application that delivers real-time risk predictions, intuitive visualizations, and actionable alerts directly to midwives.

Background & Literature survey

Recent studies demonstrate the effectiveness of machine learning models such as Random Forest, SVM, XGBoost, and deep learning architectures in predicting maternal health risks with accuracies exceeding 85%. Research by Assaduzzaman et al. (2023) and Harnal et al. (2025) highlights improved predictive performance using advanced validation techniques, while Uma Maheswari et al. (2024) emphasizes handling data imbalance and explainability using SHAP.

Despite these advancements, existing solutions lack deployment in real-time mobile applications and do not address midwives' usability needs. Furthermore, explainability is often limited to offline analysis, and no prior systems incorporate voice or haptic feedback to support rapid decision-making. The proposed system bridges this gap by embedding lightweight ML models within a mobile app, supported by interactive dashboards and multimodal alerts.

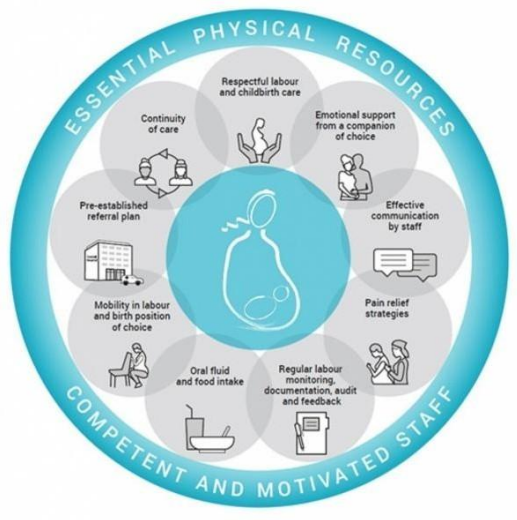

Figure 1 Midwifery Care Phases (Antenatal, Intrapartum)

Figure 2 Antenatal Monitoring Parameters (BP, FHR etc.)

Research Gap

Although machine learning techniques for maternal risk prediction are well researched, the following gaps remain:

- Lack of midwife-focused mobile applications integrating ML predictions
- Absence of real-time interactive data visualization
- No use of voice or haptic feedback for urgent alerts
- Limited cultural and language adaptation for Sri Lankan users
- Minimal usability and clinical simulation testing with real users

The proposed system uniquely addresses all these gaps.

Table 1 Research Gap

Research Paper	Research Paper Discussion	Research Gap
Assaduzzaman et al. (2023)	Built ML classifiers (SVM, Random Forest) achieving ~89% accuracy for early maternal risk prediction.	Not integrated into a midwife-focused mobile app; no real-time interface.
Harnal et al. (2025)	Compared percentage split vs. k-fold cross-validation; k-fold improved accuracy by ~5%.	Lacked interactive explainability or deployment in clinical settings.
Uma Maheswari et al. (2024)	Applied SMOTE and SHAP with XGBoost to handle imbalance and explain feature importance; recall improved by 12%.	No live dashboards or clinical workflow integration; explainability only post-hoc.
Rahman & Alam (2023)	Used CNN/LSTM models with LIME for deep learning risk prediction; achieved 91% F1-score.	High computational load; no real-time on-device inference or voice/haptic alerts.
Ravi et al. (2022)	Analyzed time-series trajectories of maternal risk, identifying distinct risk patterns over pregnancy.	Did not embed predictive models into a user interface; no mobile deployment or midwife-centered design.

Table 2 System Gap

Feature	Proposed System	Assaduzzaman et al. (2023)	Harnal et al. (2025)	Uma Maheswari et al. (2024)	Rahman & Alam (2023)	Ravi et al. (2022)
Uses machine learning	✓	✓	✓	✓	✓	✓
Includes data visualization	✓	✗	✗	✗	✗	✗
Focus on midwives	✓	✗	✗	✗	✗	✗
Includes real-time risk detection	✓	✓	✓	✓	✓	✓
Voice & Haptic feedback	✓	✗	✗	✗	✗	✗
Supports local cultural adaptation	✓	✗	✗	✗	✗	✗
Prototype tested with users	✓	✓	✗	✗	✗	✗

Research Problem

Midwives in Sri Lanka lack access to intelligent, real-time digital tools that transform maternal health data into actionable insights during antenatal care.

Problem Statement:

How effective is an interactive mobile application with machine learning and data visualization in improving midwives' decision-making in maternal care?

Objectives

Main Objectives

To develop and evaluate a machine learning–based interactive mobile application that improves antenatal risk detection and decision-making among midwives.

Specific Objectives

1. Design and integrate lightweight supervised machine learning models for real-time maternal risk prediction.
2. Develop interactive dashboards to visualize maternal health trends and risk levels.
3. Implement voice and haptic alerts to support rapid clinical response
4. Ensure cultural relevance through multilingual (Sinhala, Tamil, English) UI design.\

5. Evaluate system usability, predictive accuracy, and decision impact through midwife testing and clinical simulations

Mobile Application Development

• Cross-platform (Flutter / React Native).
• Interactive UI/UX for maternal monitoring & birth-phase modules.
• Local language support and offline accessibility.

Data Analysis & Dashboard Design

• Maternal health data stored in database (SQL/Firebase).
• Visualization of trends (BP over time, fetal growth).
• Alerts for abnormal values.
• Student training performance reports.

Methodology

The project follows a mixed-method development and evaluation approach:

1. Requirement Gathering – midwife interviews and review of FHB guidelines.
2. Data Collection & Preprocessing – anonymized maternal datasets.
3. ML Model Development – training and evaluating supervised models; conversion to TensorFlow Lite.
4. Mobile App Development – Flutter/React Native implementation with dashboards and alerts.
5. Explainability & Visualization – SHAP/LIME integration and interactive charts.
6. Testing & Validation – usability testing (SUS), clinical simulation trials, and performance evaluation.

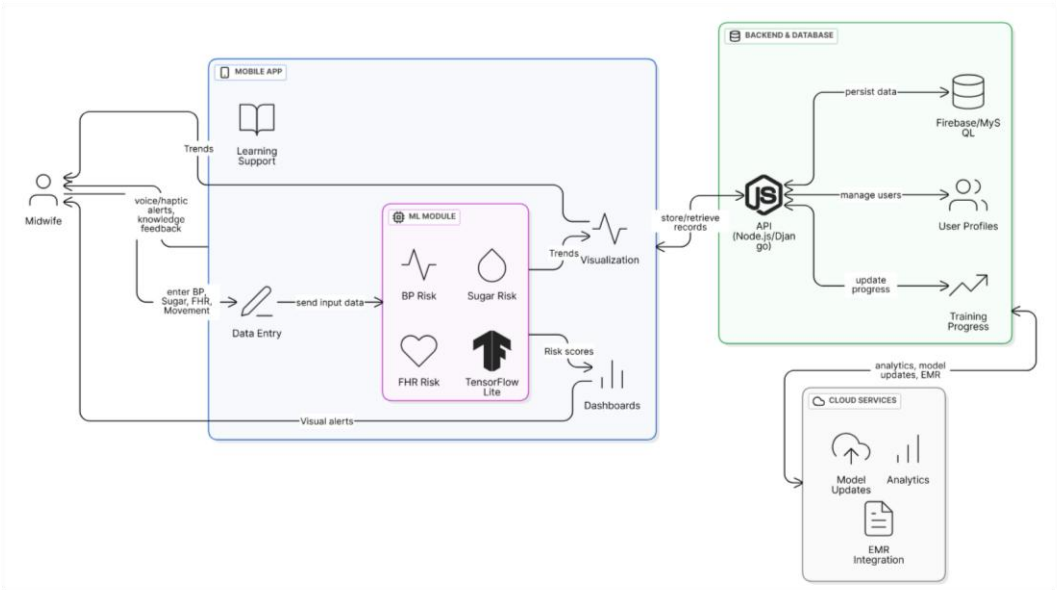

Figure 4 Individual System Diagram

Table 3 Technologies Used

Technology	Description
Flutter / React Native	Cross-platform mobile development framework used to build the mobile application (Android & iOS).
Firebase / MySQL	Backend database solutions to store maternal health data, training data, and user profiles.
RESTful API (Node.js / Django)	Middleware for communication between mobile app and backend database.
Python (Pandas, Scikit-learn)	Used for data analysis, trend prediction, and generating insights from maternal health data.
Chart.js / Power BI	Data visualization tools for creating dashboards and performance analytics.
Figma	UI/UX design tool used for prototyping the mobile application.
Android Studio / VS Code	Development environments for coding, debugging, and testing.
GitHub	Version control system for managing collaborative development.

Functional Requirements

1. Input and tracking of maternal health data (BP, FHR, blood sugar, Hb).
2. Real-time ML-based risk prediction.
3. Interactive dashboards with trend visualization.
4. Voice and haptic alerts for abnormal conditions.
5. Multi-language and offline support.

Non-Functional Requirements

1. Secure handling of sensitive health data.
2. Fast on-device ML inference.
3. High usability and accessibility.
4. Scalability for multiple clinics and users.

System Requirements

Hardware Requirement

1. Smartphones (Android/iOS).
2. Mid-range laptops for development.

Software Requirements

1. Flutter/React Native framework.
2. Firebase / SQL database.
3. Power BI / Chart.js for dashboards.

Table 4 Work Breakdown Chart

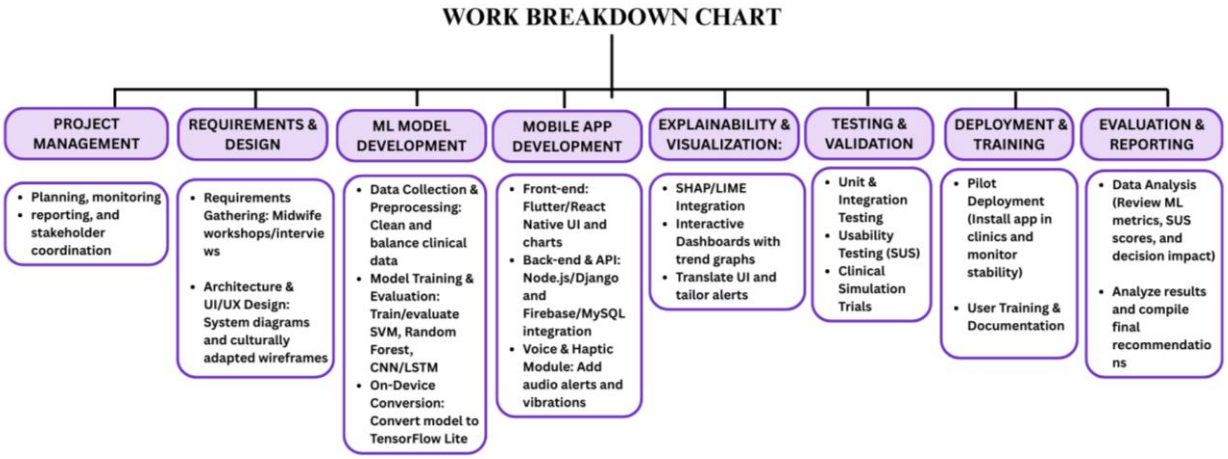

Table 5 Grantt Chart

Phase	Sep	Oct	Nov	Dec	Jan	Feb	Mar
Literature Review & Benchmarking							
AI Model Development (CVA, CI, Posture)							
Data Preprocessing & Testing (OpenCV)							
UI/UX Design (Figma, Audio Guidance)							
App Development (Flutter, SQLite)							
Offline Functionality & Localization							
Midwife Usability Testing & Feedback							
Final Validation & Play Store Prep							

Description of Personal and Facilities

Supervisors
Our supervisor Mr. Didula Thanaweera Araghchi and Co-Supervisor Mrs. Narmada Gamage, who are experts in the technical field, will provide guidance in software development and technology integration.

External Supervisor
Our external supervisor is a midwife who has been in the field for over 26 years. She shared real experiences, how she manages mothers before, during and after birth. Her knowledge helped us understand the real problems midwives face daily. With her guidance, we were able to plan the app feature more practically and usefully.

Facilities
The project is carried out with support from the Humanities and Nursing Departments of SLIIT. Access is provided to medical books, maternal health guidelines, and computer labs for app development and testing. This setup ensures that the mobile app is developed in line with real-world midwifery practices in Sri Lanka.

Sample of Midwifery Students
A group of 15–20 midwifery and nursing students will take part in the testing phase. They will use the app, give feedback on its features, and help improve its design and content. This hands-on testing also gives students practical learning opportunities.

Device Access
The app will be tested on commonly used Android smartphones, including low and mid- range devices. This ensures it works well for public health midwives in both urban and rural areas. The testing will also cover local language support, offline access, and data entry functions.

Budget and Budget Justification

Table 6 Budget Table

Expenses	
Requirement	Cost (Rs.)
Mobile Devices for Testing (Android + iOS)	180,000
Backend Server Hosting (1 year)	60,000
Firebase / Cloud Database (annual)	45,000
Software Licenses (Figma, IDEs, API tools)	30,000
Data Analysis Tools (Python Libraries, Power BI Pro)	50,000

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Digitalizing Midwifery Care: AR and VR Solutions for Maternal Education

Project Proposal Report

Mahawe Koralalage Samadini Pasanya Mahawegedara

BSc (Hons) in Information Technology – Interactive Media

Department of Information Technology

Sri Lankan Institute of Information Technology

August 2025

Digitalizing Midwifery Care: AR and VR Solutions for Maternal Education

Midwifery Training and Postnatal Care Management App

Project Proposal Report

Mahawe Koralalage Samadini Pasanya Mahawegedara

BSc (Hons) in Information Technology – Interactive Media

Department of Information Technology

Sri Lankan Institute of Information Technology

August 2025

Declaration, copyright statement and the statement of the supervisor

We declare that this is our own work and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
Mahawegedara M K S P	IT21038778	

The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Mr. Didula Thanaweera Arachchi

Signature of the supervisor: .

Date: 8/29/2025

Abstract

The report consists of a comprehensive outline of how a Virtual Reality (VR) training environment and a postnatal support mobile application can modernize and improve the maternal healthcare sector by empowering midwives in delivering quality care. This VR training module focuses on newborn handling tasks such as carrying, cleaning, and breastfeeding guidance. The environment integrates interactive 3D models of infants, a virtual clinical space, and structured training sessions, creating an immersive, repeatable simulation for practical learning. Alongside the VR module, a mobile app is developed for midwives to manage postnatal care — enabling them to digitally record mother and baby vitals, generate reports, and receive automated reminders. The system ensures improved hands-on skills, better documentation, and accessible postnatal support.

The research gap involves the lack of integrated systems that combine immersive VR training with mobile-based postnatal care tools tailored specifically for midwives. Existing studies such as the *Virtual Midwife for Pregnant Women and Alert System* focus mainly on mobile alerts and communication, while the *Mobile-based VR Learning Environment for Postpartum Hemorrhage* highlights VR training for emergency situations rather than routine newborn care. Additionally, *A User-Centered Design of Natural Language Processing for Maternal Monitoring* supports mobile maternal tracking but lacks practical task integration and midwife-specific workflows. These studies point toward the need for a more complete, connected solution addressing both training and real-world support.

The novelty of this system lies in its dual-approach design: a VR-based training module for safe, interactive skill development, and a feature-rich mobile application focused on the postnatal stage of maternal care. The VR setup replicates a real clinical environment with interactive guidance and performance tracking, while the mobile app supports Sinhala voice-to-text, digital recordkeeping, and cloud-synced reports. References from IEEE research strengthen the foundation of this work, combining immersive learning and digital health tools into a unified system. Altogether, the project highlights the vital role of technology in advancing maternal care and midwifery education.

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List of Abbreviations

Abbreviation	Description
VR	Virtual Reality
AR	Augmented Reality
3D	3 Dimensional
EMS	Emergency Medical Service
API	Application Programming Interface
UI	User Interface
UX	User Experience
VS	Visual Studio
SDK	Software Development Kit
NHK	National Hospital Kandy

Introduction

Background & Literature survey

A fundamental competency in maternal and pediatric healthcare, particularly for midwives during and after childbirth, is newborn handling. These abilities include proper cleaning techniques, safe infant carrying techniques, nursing advice, and fundamental postnatal evaluations. During their clinical and academic training, midwifery students are expected to become proficient in these duties. However, many students lack sufficient practical experience prior to becoming certified experts because of restricted exposure to real-life situations and safety issues when handling babies.

Techniques like supporting the head and neck, washing without causing discomfort, and positioning for nursing are all part of proper infant handling. Every movement calls for attention and accuracy. Real-world conditions must be replicated in training settings so that midwives can learn by making mistakes and fixing them without endangering live babies. By providing trainees with a virtual 3D infant, a realistic clinical setting, and guided feedback, interactive virtual reality technology improves skill development through organized courses. Students can safely and regularly practice genuine events while interacting with lifelike models because to the immersive nature.

Figure 1: Safe Newborn Handling Positions in VR

After receiving their certification, midwives are in charge of managing and recording postpartum care. Manual data entry and conventional paper-based methods take a lot of time and are prone to mistakes. A secure mobile application designed especially for midwives is created to address this. With the use of this government-integrated program that allows for Sinhala voice-to-text input, midwives can easily collect mother and infant data. Smart vaccine and clinic appointment reminders, automated daily, monthly, and annual report generation, and a data analytics dashboard to monitor trends like average newborn weight by age are all included in the app. Previous studies have examined mobile health solutions, such as the Virtual Midwife Alert System [1] and the NLP-Based Maternal Monitoring System [2], but they have not integrated midwife-specific training and postnatal care elements.

In order to support the care of mothers and newborns, this study focuses on two different but complimentary improvements. As part of their pre-professional education, student midwives can safely practice critical infant handling methods in an immersive and interactive virtual reality (VR) training session. The second is a mobile application designed specifically for qualified midwives to handle the postpartum care period. It provides real-time analytics, smart notifications, automatic reporting, and secure voice-to-text data entry. When combined, these developments seek to enhance postnatal care quality and training efficacy in a contemporary healthcare setting.

Research Gap

Several research papers were reviewed in the domain of maternal and newborn care using VR and mobile-based applications. While all selected studies offered valuable insight into maternal health technologies, only a few directly addressed postnatal care tools for midwives or VR-based training for newborn handling. Most of the existing systems were either mobile-only with basic features or focused on emergency care rather than routine postnatal responsibilities.

Research A, "Virtual Midwife for Pregnant Women and Alert System", focuses on developing a mobile application for communication and emergency alerts during pregnancy. Although it supports general monitoring and notifications, it lacks dedicated features for structured postnatal care workflows, such as digital health tracking, data analytics, and official report generation used by midwives in the after-birth phase.

Research B, *"Mobile-based Virtual Reality Learning Environment for Postpartum Hemorrhage Management Training"*, presents a VR solution for training on maternal emergencies. It shows how VR can improve healthcare skills but does not cover routine

newborn care training or tasks that midwives must learn before certification, such as handling and breastfeeding guidance in the postnatal stage.

Research C, *"A User-Centered Design of NLP for Maternal Monitoring Chatbot System"*, explores the use of intelligent systems for maternal tracking through mobile chatbots. While it introduces useful voice-based interaction, it lacks Sinhala voice-to-text integration, field-level data analytics, and secure record-keeping designed for official midwife use. It is not aligned with the workflow standards required for postnatal care and reporting.

Table 1 Research Gap

Application Reference	Postnatal Workflow Coverage	VR Newborn Handling Training	Sinhala Voice-to-Text	Secure Recordkeeping	Data Analytics	Report Generation
Research A	x	x	x	x	x	x
Research B	x	✓	x	x	x	x
Research B	x	x	x	x	x	x
Our Proposed System	✓	✓	✓	✓	✓	✓

Research Problem

Learning how to handle a newborn safely can be challenging for midwifery students, especially during their training period where live practice opportunities are limited. Tasks like carrying, cleaning, and positioning a baby for breastfeeding require precision and confidence, but incorrect technique can pose risks. Without hands-on exposure, many students enter the workforce without fully developing these essential skills.

To address this, the proposed VR-based newborn care training environment gives midwifery students a safe and repeatable way to learn. The system offers interactive guidance on core postnatal tasks using realistic 3D newborn models and a virtual clinical setup. It simulates common scenarios midwives will face and allows them to practice with feedback, building real-world readiness through immersive repetition. Tasks such as holding the baby, supporting the neck, or positioning for breastfeeding are covered in structured modules.

On the other hand, practicing midwives face the challenge of keeping consistent digital records and managing after-birth care across many mothers. Manual data entry is time-consuming and increases the risk of missing critical follow-ups like vaccinations or clinic visits. Current systems also lack secure integration with local workflows and languages.

To solve this, a dedicated mobile application is being developed for midwives to manage after-birth care efficiently. The app supports Sinhala voice-to-text data entry, smart notifications for vaccine and clinic schedules, automated report generation, and data analytics (e.g., average child weight per age). It is designed with future government integration in mind, ensuring it can scale securely and support public health infrastructure. Together, both components target the real gaps in training and postnatal care management faced by midwives today.

Objectives

Main Objectives

The main objective of this project is to develop a realistic VR training environment for midwifery students to practice essential newborn handling techniques, and to build a secure mobile application for certified midwives to manage postnatal care efficiently.

VR Training Module : After wearing the VR headset and controllers, the midwifery student is placed in a virtual clinical room where various newborn care tasks can be practiced. The student can navigate to modules such as safe baby handling, cleaning, or breastfeeding assistance. Each task is guided through interactive prompts using a 3D animated newborn, allowing the learner to develop real-world skills in a safe, immersive space. Throughout the training, the student receives visual and audio feedback, ensuring mistakes can be corrected in real time and the learning is repeatable. The goal is to build confidence and accuracy before clinical placement.

Midwife Mobile App (After-Birth Phase) : The second component of this project is a mobile application designed specifically for government-authorized midwives. This app allows midwives to securely record mother and baby health information using Sinhala voice-to-text, minimizing the need for manual typing. The app sends automated reminders for vaccinations and clinic visits, and generates daily, monthly, and yearly reports of the midwife's activity. A data analytics dashboard helps track trends such as average infant weight across age groups. While not yet government-integrated, the system is built with the security and features required for future collaboration with public healthcare bodies.

Specific Objectives

VR Integration and Simulation Development

A fully immersive VR training environment will be developed using Unity Engine and OpenXR, enabling realistic simulation of newborn care procedures. This includes integrating interactive elements such as hand tracking and object interaction for tasks like carrying, cleaning, and positioning a newborn. The VR system serves as the core of the student training module; if this integration fails, the entire immersive learning experience would be compromised, making it the most critical technical objective.

3D Newborn Models and Clinical Environment Design

Realistic 3D models of newborns and a clinical postnatal care environment will be created to simulate hands-on procedures. Tools such as Blender or Maya will be used to model the virtual environment, which will include items like a hospital bed, infant cleaning station, and breastfeeding setup. Anatomical accuracy is essential, so references from maternal health research and neonatal care standards will guide the design. The objective

is to ensure realism so students can practice and build confidence in a safe yet life-like virtual setting.

VR User Interface (UI) and Task Flow Integration

An intuitive UI will be designed using Unity's VR toolkit to allow students to navigate through training modules, tutorials, and practice sessions. The system will include guided tasks and feedback-based challenges to reinforce learning. Voice prompts, interactive menus, and visual indicators will be embedded to help learners move through the training efficiently. Each task will track completion and errors, forming the foundation for assessment.

Mobile Application Development (After-Birth Phase)

A mobile app tailored for certified midwives will be developed using Flutter or Android Studio, focusing on postnatal care data management. This app will include secure login, Sinhala voice-to-text support, and cloud-based data storage. It will allow midwives to log mother and child information, send vaccination and clinic visit reminders, and auto-generate daily, monthly, and annual reports. Data analytics tools will also be embedded to track health trends like average infant weight by age group.

Data Security and Future Integration Readiness

Since the midwife app handles sensitive patient data, implementing high-level data security protocols is essential. This includes encrypted databases, role-based access controls, and compliance with health data protection standards. The system will be developed with a scalable architecture to support potential future integration with government health systems, pending official approval.

Methodology

The proposed system consists of two key components: a VR-based newborn care training module and a mobile application for certified midwives. Each component is developed using a combination of modeling tools, mobile frameworks, and immersive simulation engines to ensure a practical, user-friendly, and efficient experience for both student midwives and professional practitioners.

To begin with, the VR training module is developed using the Unity Engine alongside OpenXR for VR integration. This forms the foundation for immersive interaction. The clinical environment and newborn handling tasks are created using Blender, referencing real-world midwifery training practices. The 3D newborn models, clinical tools, and the simulation space are imported into Unity, where C# is used to program interactive controls, training tasks, and feedback systems. The result is a fully interactive learning lab

where students can perform safe handling, cleaning, and breastfeeding positioning in a risk-free setting.

The mobile application is developed separately using Flutter or Android Studio and is targeted for official use by government-approved midwives. The app includes a secure login, voice-to-text support in Sinhala, and allows the recording of mother and baby health information. Smart notifications are integrated to remind midwives of important tasks such as upcoming vaccinations or clinic checkups. The app generates daily, monthly, and annual activity reports and includes a data analytics dashboard that highlights trends, such as average infant weight across age ranges. Cloud database support is included for data backup and scalability.

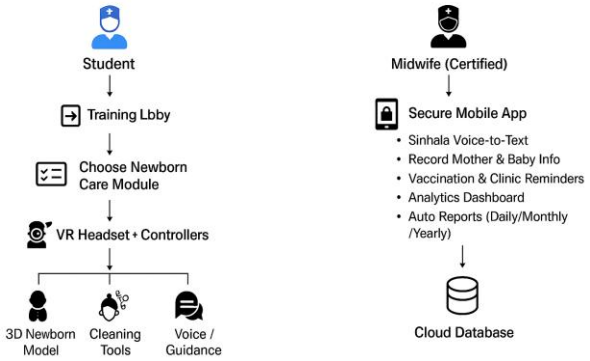

Figure 2: Individual System Diagram

Figure 2 shows the Individual System Diagram of the proposed solution, which includes both the VR Newborn Care Training Module and the Midwife Mobile Application. Initially, the student midwife is placed in a virtual training lobby or learning environment where they can select from a set of newborn care training modules. Once a module is selected, the student uses a VR headset running on Unity Engine with OpenXR integration, enabling them to interact with 3D newborn models, tools, and guided voice-based instructions in a realistic virtual environment.

On the other hand, the certified midwife accesses a secure mobile application designed for the after-birth care phase. Through this app, the midwife can record mother and baby information using Sinhala voice-to-text, set clinic and vaccination reminders, and generate automated daily, monthly, and yearly reports. All recorded data is synced to a cloud database, ensuring data integrity, analytics, and accessibility. Both systems are independent but serve a shared goal of improving maternal and newborn care.

The main component the VR environment consists of 5 different features or requirements, 3D models of equipment, and 3D heart model, various heart beat sounds, VR player control that gives the student intractability, and finally training, challenges and tasks all helping the student to train and learn using the environment. These all 5 features need a dataset of Cardiac Auscultation sounds, this includes various heart beat sounds and list of conditions that these sounds fit into.

Table 2 Technologies Used

Technology	**Description**
Unity Engine	Core development platform for VR training environment
OpenXR SDK	Used for VR hardware compatibility and immersive interaction
C# Language	Backend scripting and interaction logic in Unity
Blender / Maya	Tool for creating 3D newborn models and clinical environment
Flutter / Android Studio	Frameworks for building the mobile app for midwives
Firebase / Cloud DB	Cloud-based data storage and backup for mobile app
Google Speech API	For Sinhala voice-to-text conversion in mobile app
Visual Studio Code	Code editing and debugging environment

Functional Requirements

1. The VR training module should allow the user (midwifery student) to fully interact with the virtual environment, including selecting training modules and manipulating 3D newborn models.
2. Tasks such as safe carrying, cleaning, and breastfeeding guidance should function accurately, along with guided instructions and feedback mechanisms.
3. The mobile application should allow certified midwives to record data using Sinhala voice-to-text, send smart reminders, and generate automated reports.

Non-Functional Requirements

1. Medical Accuracy – All training tasks and postnatal care workflows must reflect up-to-date, verified medical procedures.

2. Model Accuracy – 3D newborn and tool models should be designed with realistic proportions and detail for effective training.
3. System Security – The mobile app should maintain a high level of data security to protect sensitive patient information.
4. User-Friendliness – Both the VR system and mobile app should offer an intuitive user interface for smooth navigation and ease of use.
5. Performance – The app and VR system should load quickly and operate smoothly without lag during training or data entry.

System Requirements

The VR module should include realistic audio-visual feedback (e.g., newborn responses, voice instructions), structured training modules, and performance tracking.
The mobile app must support real-time data recording, automated reminders, cloud syncing, and a dashboard for analytics and reporting.
Both components must operate independently but be designed for scalability and maintainability.

Hardware Requirements

- VR Headset (e.g., Oculus Quest, HTC Vive) with motion controllers
- High-performance PC capable of running Unity VR environments
- Smartphone or tablet for running the mobile midwife app
- Stable internet connection for cloud-based data syncing

.

Software Requirements

- Unity Engine for developing the VR training module
- OpenXR SDK for VR device compatibility
- Blender / Maya / 3ds Max for modeling 3D newborns and clinical environments
- Flutter / Android Studio for mobile app development
- Firebase or equivalent for real-time database and cloud storage
- Google Speech API for Sinhala voice-to-text functionality

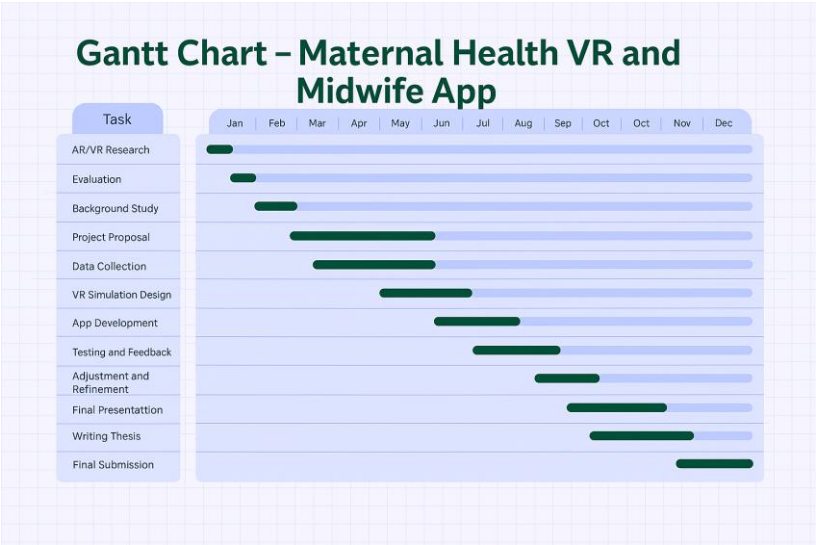

Figure 3 Gantt Chart

Description of Personal and Facilities

Supervisors

Our supervisor Mr. Didula Thanaweera Arachchi and Co-Supervisor Ms. Narmada Gamage, who are experts in the technical field will provide guidance in software development and technology integration.

External Supervisor

Guiding the healthcare aspect of the project is Senior Midwife Sandya Liyanage, who brings over 20 years of experience in maternal and postnatal care. Her extensive field expertise ensures the accuracy and practical relevance of both the VR training scenarios and the mobile application's features. Her input has been invaluable in shaping realistic training modules, improving usability for midwives, and validating clinical workflows used in the after-birth care phase.

Facilities

This research project is conducted in collaboration with the Humanities Department of SLIIT, with active involvement from nursing students. Through this partnership, access is provided to medical resources, nursing textbooks, and clinical guidelines. These facilities and references ensure that the developed VR training environment closely aligns with real-world midwifery practices, enhancing the clinical relevance and accuracy of the system

Sample of Medical Students and midwives

The project involves a sample group of medical and nursing students who will test the VR training module and provide detailed feedback. This offers them a valuable hands-on learning experience while contributing to system improvement. In addition, a group of certified midwives will test the mobile application, focusing on postnatal care tasks such as voice-to-text data entry, reminders, and report generation. Their feedback will be critical to ensuring the app meets real-world midwifery needs and follows practical workflows used in community healthcare settings.

High-Quality VR Equipment

For students to experience the environment as well as for testing purposes a high-quality VR headset and controllers are needed. These VR devices enables students to interact with the environment. Using a high-quality modern VR equipment provides access to the latest technology for medical purposes and simulation training.

Budget and Budget Justification

Table 3 Budget Table

Expenses	
Requirement	Cost (Rs.)
VR Meta Quest 3	186,000
Unity Engine License	57,000/month
Graphics Card (Nvidia RTX 3050)	97,500
Mobile Phone (Google Pixel 8 Pro)	200,000
Total Cost	540,500

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